

PROJEXIS

Web-Based Research & Development Management System

Role-Based Process Guide & System Specification

Comprehensive documentation covering Superadmin, Admin, Research Adviser, and Student roles — detailing responsibilities, workflows, permissions, and system limitations.

★ SUPERADMIN

◆ ADMIN

▲ RESEARCH ADVISER

● STUDENT

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1. System Overview

1.1 Purpose & Scope

Projexis is a multi-tenant, web-based Research and Development Management System designed to streamline the entire lifecycle of academic and institutional research — from project inception through documentation, collaboration, and final delivery. The platform supports four distinct roles operating across isolated department tenants, ensuring data privacy, clear authority chains, and efficient coordination between administrators, research advisers, and student researchers.

1.2 System Architecture & Multi-Tenancy

Projexis operates on a multi-tenant architecture where each Department functions as an independent tenant. A single Superadmin governs the entire platform and provisions departments. Each department has exactly one Admin who manages all operations within that tenant boundary. Data between departments is strictly isolated — no cross-department data leakage occurs.

Tier	Role	Scope	Count
Platform	Superadmin	Entire Projexis instance	1 (global)
Department	Admin	Single department tenant	1 per dept.
Group	Research Adviser	Assigned research groups	Many per dept.
Group	Student	Own group only	Many per group

1.3 Nine Core Subsystems

1. User Management Module	Manages accounts, roles, groups, and authentication for all tiers.
2. Project Tracking Module	Tracks research projects, milestones, statuses, and progress timelines.
3. Product Development Lifecycle Module	Manages phases from ideation → development → testing → deployment.
4. Research Documentation Module	Central repository for proposals, manuscripts, and deliverable files.
5. Collaboration Tools Module	Messaging, commenting, file sharing, and group communication channels.
6. Innovation Analytics Module	Insights into project velocity, output quality, and innovation KPIs.
7. Dashboard & Reporting Module	Role-tailored dashboards with real-time statistics and exportable reports.
8. Notification System	Automated in-app and email alerts for deadlines, submissions, and updates.
9. Audit Logs	Immutable event logs for all system actions — who did what and when.

2. Role Hierarchy & Access Model

2.1 Role Definitions

★ Superadmin	The platform owner. Has unrestricted access to all system settings, department provisioning, and admin account management. Cannot be created by any other role — must exist in the system by default.
◆ Admin	Operates within a single department tenant. Manages research groups, assigns advisers, configures projects, and oversees all department operations. Created exclusively by the Superadmin.
▲ Research Adviser	Assigned to one or more student research groups by the Admin. Manages deliverables, requirements, deadlines, and project statuses for their assigned groups. Created by the Admin.
● Student	Member of a research group. Can view announcements, track deliverables, submit documents, and communicate within their group. Created by the Admin and assigned to a group.

2.2 Permissions Matrix

Legend: ✓ Full = full access | ✓ Dept = department scope | ✓ Own = own records only | ✓ View = read-only | ✗ = no access

Module / Feature	Superadmin	Admin	Adviser	Student
Department Management	✓ Full	✓ Dept	✗	✗
Admin Account CRUD	✓ Full	✗	✗	✗
Adviser Account CRUD	✗	✓ Dept	✗	✗
Student Account CRUD	✗	✓ Dept	✗	✗
Group Creation & Assignment	✗	✓ Dept	✗	✗
Assign Adviser to Group	✗	✓ Dept	✗	✗
Project Creation	✗	✓ Dept	✓ Own	✗
Deliverable Management	✗	✓ Dept	✓ Own	✓ View
Status Updates (Deliverables)	✗	✓ Dept	✓ Own	✗
Document Submission	✗	✗	✗	✓ Own
Announcement Publishing	✗	✓ Dept	✓ Own	✓ View
Collaboration / Messaging	✗	✓ Dept	✓ Own	✓ Own
Reports & Analytics	✓ Full	✓ Dept	✓ Own	✗

System Configurations	✓ Full	✓ Dept	✗	✗
Audit Logs	✓ Full	✓ Dept	✗	✗
Notification Settings	✓ Full	✓ Dept	✓ Own	✓ Own
Dashboard Access	✓ Full	✓ Dept	✓ Own	✓ View

★ SUPERADMIN

3. Superadmin — Full Process Guide

The Superadmin is the highest-authority user in the Projexis platform. This role is responsible for the structural integrity of the entire system — from provisioning departments to maintaining compliance through audit logs. There is exactly one Superadmin account per Projexis instance.

3.1 Superadmin Onboarding Workflow

#	Step / Action	Outcome
1	Log in with platform root credentials	Superadmin dashboard loaded
2	Complete platform configuration (name, timezone, branding)	Platform identity set
3	Set global password policies and session rules	Security baseline established
4	Configure email/SMTP notification settings	Notification engine active
5	Enable or disable available system modules	Module visibility configured
6	Review and finalize audit log retention policies	Compliance framework ready

3.2 Department Management

Departments are the top-level organizational units and serve as tenant boundaries in Projexis. Each department operates in isolation from others.

#	Step / Action	Outcome
1	Navigate to Department Management → Create New Department	Department provisioning form opened
2	Fill in: Department Name, Code, Description, Academic Year	Tenant identity defined
3	Set department status (Active / Inactive / Archived)	Tenant lifecycle state set
4	Save — department tenant created and isolated	New department available in directory
5	Repeat process for each institutional department	All tenants provisioned
6	Edit department details when needed (rename, update info)	Department record updated
7	Archive department if discontinued (data preserved)	Department moved to archive

3.3 Admin Account Management

Only the Superadmin can create Admin accounts. Each Admin is bound to one department.

#	Step / Action	Outcome
1	Navigate to User Management → Admins → Create Admin	Admin creation form opened
2	Input: Full Name, Email, Department assignment, Temporary Password	Admin account form filled
3	Assign the Admin to the target department	Admin-department binding set
4	Set account status (Active) and send credential email	Admin receives login credentials
5	Monitor admin activity via Audit Logs	Admin activity tracked
6	Deactivate/suspend Admin account if needed	Admin access revoked without data loss
7	Reassign department to a new Admin if current one is removed	Continuity maintained

3.4 Platform Configuration

Platform Settings Controlled by Superadmin
• Set platform name, logo, and color theme (white-labeling) • Configure academic calendar and semester/term cycles • Manage system-wide announcement broadcasts visible to all tenants • Control feature flags — enable/disable modules globally • Set file upload limits and accepted document formats • Configure backup schedules and data retention policies • Manage third-party integrations (Google Drive, email SMTP, etc.)

3.5 Audit & Compliance

The Superadmin has access to the complete immutable audit trail of the entire platform.

- View all system events: logins, logouts, data changes, deletions
- Filter logs by role, department, date range, and action type
- Export audit logs to CSV/PDF for compliance reporting
- Receive alerts for suspicious activity (failed logins, mass deletions)
- Cannot alter or delete audit log entries (immutable by design)

3.6 Superadmin — Limitations & Restrictions

- ✗ Cannot manage research projects, deliverables, or student submissions directly
- ✗ Cannot view personal student or adviser data within a department (privacy boundary)
- ✗ Cannot create Research Adviser or Student accounts (delegated to Admin)
- ✗ Cannot be created, edited, or deleted by any other user role
- ✗ Cannot bypass audit log immutability — all actions are permanently logged
- ✗ Cannot access department-internal messaging or group collaboration spaces
- ✗ Limited to one Superadmin account per platform instance

◆ ADMIN

4. Admin — Full Process Guide

The Admin is the primary operational manager within a department. They are responsible for setting up and managing all users, research groups, projects, and communications within their assigned department. The Admin acts as the bridge between platform governance (Superadmin) and day-to-day research operations.

4.1 Department Setup Workflow

#	Step / Action	Outcome
1	Log in with Admin credentials received from Superadmin	Admin dashboard loaded
2	Complete department profile (description, academic period, logo)	Dept profile configured
3	Set department-level notification preferences	Notification channels active
4	Configure grading schemes and evaluation rubrics	Assessment framework ready
5	Set academic calendar milestones and deadlines	Timeline baseline established
6	Create category tags for research types/domains	Taxonomy defined

4.2 User Management — Advisers & Student Groups

Creating Research Adviser Accounts

#	Step / Action	Outcome
1	Navigate to Users → Research Advisers → Add Adviser	Adviser form opened
2	Enter: Full Name, Email, Specialization, Credentials	Profile information set
3	Assign initial groups (optional — can be done later)	Adviser-group links configured
4	Set account active and dispatch credentials via email	Adviser receives login email
5	Adviser appears in the department directory	Adviser is now assignable to groups

Creating Student Research Groups

#	Step / Action	Outcome
1	Navigate to Groups → Create New Group	Group creation form opened
2	Enter Group Name, Research Title (tentative), Academic Year, Semester	Group identity set
3	Add students: enter each student's name, email, student ID	Student accounts created and linked
4	Assign a Research Adviser to the group	Adviser-group binding established
5	Set group status: Active / On Hold / Completed	Group lifecycle state set
6	Notify adviser and students via system notification	All members receive onboarding alert
7	Students and adviser now see the group on their dashboards	Group is operational

4.3 Project & Research Management

Admins can create, oversee, and monitor all research projects within their department.

#	Step / Action	Outcome
1	Navigate to Projects → Create Research Project	Project form opened
2	Enter: Title, Abstract, Research Type, Start Date, Target End Date	Project metadata set
3	Link project to a student group	Project assigned to a group
4	Define project phases (Proposal → Development → Final Defense)	Phase structure created
5	Configure evaluation criteria per phase	Assessment criteria linked
6	Monitor overall project progress from department dashboard	Real-time tracking active
7	Archive or close project upon completion	Project archived with records

4.4 Resource & Schedule Management

Admin Resource Controls

- Manage department resource library (templates, guidelines, references)
- Schedule and book rooms or facilities for research presentations
- Assign evaluation panels to final defense schedules
- Upload and publish department-wide announcement broadcasts

- Create and manage department-wide events on the academic calendar
- Track resource utilization and flag over-allocated advisers
- Approve or reject resource requests from research groups

4.5 Reporting & Analytics

- View department-level project completion rates and submission timelines
- Generate per-group progress reports with deliverable status breakdowns
- Export reports to PDF or Excel for institutional submissions
- View adviser workload metrics (number of groups, pending reviews)
- Monitor overdue deliverables and escalate notifications automatically
- Access innovation analytics: research output volume, quality trends

4.6 Admin — Limitations & Restrictions

- ✗ Cannot access or modify data belonging to other departments
- ✗ Cannot create or modify Superadmin accounts
- ✗ Cannot alter immutable audit log records
- ✗ Cannot override system-level configurations set by Superadmin
- ✗ Cannot directly submit deliverables on behalf of students
- ✗ Cannot add, edit, or delete system modules — only configure within scope
- ✗ Cannot access internal group messaging unless explicitly permitted
- ✗ Deactivation of Admin account must be done by Superadmin only

▲ RESEARCH ADVISER

5. Research Adviser — Full Process Guide

The Research Adviser is the academic mentor and operational supervisor of one or more student research groups. They manage the day-to-day execution of the research project — defining deliverables, setting requirements, updating statuses, providing feedback, and guiding students from proposal to completion.

5.1 Onboarding & Profile Setup

#	Step / Action	Outcome
1	Receive credentials email from Admin	Account ready for first login
2	Log in → prompted to change temporary password	Secure password set
3	Complete profile: Bio, Specialization, Credentials, Profile Photo	Professional profile complete
4	Review assigned groups on the Adviser Dashboard	Group assignments visible
5	Accept group assignments to confirm advisory role	Groups actively managed
6	Set personal notification preferences (email + in-app)	Alerts configured

5.2 Group Management

An adviser can be assigned to multiple groups and manages each group's research progress independently.

#	Step / Action	Outcome
1	View all assigned groups from dashboard	Group list with statuses displayed
2	Open a group to see its project details, members, and timeline	Group detail view loaded
3	Post a welcome announcement to the group	Students notified of adviser's presence
4	Review existing project title and abstract (if Admin pre-filled)	Research context understood
5	Update group status as needed (Active / Needs Attention / etc.)	Status reflects current reality

5.3 Deliverable & Requirement Management

This is the core responsibility of the Research Adviser — defining what students need to do and by when.

#	Step / Action	Outcome
1	Navigate to Group → Deliverables → Add Deliverable	Deliverable form opened
2	Enter: Deliverable Title, Description, Type (Document/Presentation/Other)	Deliverable defined
3	Set Due Date and attach any reference templates/guidelines	Timeline and resource set
4	Set initial status: Pending	Deliverable published to student dashboard
5	Students receive immediate notification of new deliverable	Students are alerted
6	Review student submission when filed	Submission reviewed
7	Update deliverable status: Submitted → Under Review → Approved / Needs Revision	Status progresses
8	Add feedback/comments if revision is required	Revision instructions delivered
9	Mark as Done/Approved once satisfactorily completed	Deliverable closed
10	Add new deliverables iteratively as the project progresses	Running task list maintained

Deliverable Status Lifecycle

Status	Meaning	Who Sets It
Pending	Deliverable created, not yet submitted by students	Adviser (auto on creation)
Submitted	Students have filed the required output	System (on student submission)
Under Review	Adviser is currently reviewing the submission	Adviser
Needs Revision	Submission requires corrections before approval	Adviser
Approved / Done	Deliverable fully accepted and completed	Adviser
Overdue	Past due date and not yet submitted	System (auto-flagged)

5.4 Project Status Tracking

- Update overall project phase (Proposal / In Progress / For Defense / Completed)
- Log progress notes for each group session or milestone meeting
- Flag a group as 'Needs Attention' to alert the Admin
- View a timeline/Gantt-style view of all deliverable due dates

- See at-a-glance which deliverables are overdue, submitted, or approved
- Mark a project phase as complete to unlock the next phase

5.5 Collaboration & Feedback

Collaboration Capabilities

- Post announcements visible to all group members
- Comment directly on submitted deliverables with rich-text feedback
- Upload reference materials, templates, and sample documents for students
- Use group messaging channel to communicate with all students at once
- Schedule group meetings and log meeting minutes in the system
- Tag specific students in comments for targeted feedback

5.6 Research Adviser — Limitations & Restrictions

- ✗ Cannot access groups not assigned to them by the Admin
- ✗ Cannot create or delete Admin or student accounts
- ✗ Cannot modify department-wide settings or configurations
- ✗ Cannot create new research projects — only manages existing ones assigned by Admin
- ✗ Cannot approve or reject other advisers' group deliverables
- ✗ Cannot access the Superadmin or Admin dashboards
- ✗ Cannot alter audit logs or system notifications globally
- ✗ Cannot remove students from a group — only Admin has that authority

● STUDENT

6. Student — Full Process Guide

Students are the primary research producers in Projexis. They operate exclusively within their assigned research group, interacting with deliverables assigned by their Research Adviser, submitting documents, viewing announcements, and communicating through the group portal. The Student portal is designed to be intuitive, deadline-focused, and collaboration-friendly.

6.1 Account Activation

#	Step / Action	Outcome
1	Receive credentials email from Admin after group creation	Login credentials in inbox
2	Click activation link in email or log in directly	Account activated
3	Set a new secure password on first login	Password secured
4	Complete student profile: Name, Student ID, Program/Course, Photo	Profile complete
5	View Group Dashboard — see group name, adviser, members, project title	Fully onboarded
6	Review any existing announcements and pending deliverables	Aware of current tasks

6.2 Dashboard & Announcements

The Student Dashboard is the central hub — designed to give immediate clarity on what needs to be done, when it is due, and what has already been accomplished.

- Dashboard home shows: active deliverables, upcoming due dates, recent announcements
- Notification feed shows all new announcements from adviser and admin
- Announcements are sorted by recency; unread items are highlighted
- View the research group's full member list and adviser profile
- See the project title, abstract, and current phase at a glance
- View a personal task completion percentage for the current semester

6.3 Deliverables & Task Management

Deliverables are the core to-do items for students. The system makes it easy to see exactly what is needed, when it is due, and what the current status is.

#	Step / Action	Outcome
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1	Open Deliverables tab from group dashboard	Full deliverables list displayed
2	Each deliverable shows: Title, Description, Due Date, Status, Adviser Notes	Full context visible
3	Sort/filter by status (Pending / Overdue / Submitted / Approved)	Prioritized view
4	Click on a deliverable to see full instructions and attached templates	Detailed view opened
5	Note the due date and set a personal reminder if needed	Personal planning done
6	Prepare the required output following adviser instructions	Work in progress

6.4 Document Submission

Students submit their deliverables directly through the system — no external email required.

#	Step / Action	Outcome
1	Open the target deliverable → click 'Submit'	Submission form opened
2	Upload the required file(s) (PDF, DOCX, PPTX, or other allowed formats)	Files attached
3	Add a submission note (optional but recommended)	Context for adviser added
4	Confirm submission — system timestamps the submission	Submission recorded
5	Status automatically updates to 'Submitted'	Adviser is notified immediately
6	Await adviser review — status will change to 'Under Review'	Waiting for feedback
7	If 'Needs Revision': read adviser comments, revise, and re-submit	Revision loop starts
8	Once 'Approved': deliverable is closed — move on to the next task	Task completed

6.5 Communication

Student Communication Features

- Use the Group Chat to message all group members and the adviser simultaneously
- Tag specific group members in messages for direct communication

- View and respond to comments left by the adviser on specific deliverables
- Receive real-time notifications for: new deliverables, status changes, announcements, and messages
- Access shared resources uploaded by the adviser (templates, sample documents)
- View scheduled meeting dates posted by the adviser in the calendar

6.6 Student — Limitations & Restrictions

- ✗ Cannot view other groups' data, deliverables, or submissions
- ✗ Cannot edit or delete deliverables — only view and submit
- ✗ Cannot change deliverable status — only the adviser can update statuses
- ✗ Cannot post announcements — only view announcements from adviser or admin
- ✗ Cannot access any administrative settings or dashboards
- ✗ Cannot remove themselves from the group — must request Admin action
- ✗ Cannot view other students' submission files (individual privacy)
- ✗ Cannot communicate with advisers or admins outside the platform messaging system
- ✗ Cannot request a new adviser or change group assignment directly
- ✗ File submissions are locked after the deadline unless adviser grants an extension

7. End-to-End System Workflow

The following illustrates the complete flow from platform setup to research project completion, showing how each role hands off to the next in sequence.

SUPERADMIN

1. Creates Projexis platform instance and configures settings
2. Creates Department (e.g., 'College of Computing Studies')
3. Creates Admin account for the department
4. Sends credentials to Admin

ADMIN

5. Logs in and configures department profile
6. Creates Research Adviser accounts
7. Creates student group (e.g., 'Group Alpha — Thesis 2025')
8. Adds students to the group
9. Assigns Research Adviser to the group
10. Publishes welcome announcement

RESEARCH ADVISER

11. Logs in and reviews assigned group
12. Accepts group assignment
13. Posts introductory announcement
14. Creates deliverables: 'Chapter 1 Draft', 'Methodology Outline', etc.
15. Sets due dates and attaches templates
16. Reviews student submissions and updates statuses
17. Provides feedback; requests revisions if needed
18. Marks deliverables as Done when approved
19. Updates overall project phase (e.g., 'Proposal Approved')

STUDENT

20. Logs in and activates account
21. Views Group Dashboard — sees deliverables and due dates
22. Reads announcements from adviser and admin
23. Works on required deliverables
24. Submits files through the system before due dates
25. Reads adviser feedback and revises if needed

- 26. Re-submits revised outputs
- 27. Monitors progress toward project completion

8. Subsystem Reference

1. User Management Module

Superadmin	Create/edit/delete departments and admin accounts; manage global users
Admin	Create/manage adviser and student accounts; configure group memberships
Adviser	View own profile; update personal information
Student	Activate account; update profile information

2. Project Tracking Module

Superadmin	View all projects platform-wide; monitor completion rates
Admin	Create, assign, and monitor all projects in the department
Adviser	Track deliverable statuses and update project phase for assigned groups
Student	View project details, current phase, and personal deliverable list

3. Product Development Lifecycle Module

Superadmin	Monitor phase completion rates across all departments
Admin	Define phase templates and evaluation criteria per project
Adviser	Manage phase transitions; approve/reject phase completion
Student	View current phase and tasks required to advance

4. Research Documentation Module

Superadmin	Access archived research outputs for institutional records
Admin	Configure document categories and required file formats
Adviser	Upload reference materials; review and annotate student submissions
Student	Submit research documents; access shared templates and references

5. Collaboration Tools Module

Superadmin	Configure platform-wide collaboration settings
Admin	Monitor group communications; publish department announcements
Adviser	Post announcements; comment on deliverables; group messaging
Student	View announcements; group messaging; reply to adviser comments

6. Innovation Analytics Module

Superadmin	Platform-wide analytics: research output volume, quality trends
Admin	Department analytics: completion rates, adviser workloads, group performance
Adviser	Group-level analytics: delivery timeliness, revision rates
Student	Personal completion stats only (no department-wide data)

7. Dashboard & Reporting Module

Superadmin	Global dashboard: all departments, admins, active projects
Admin	Department dashboard: groups, advisers, project statuses, reports
Adviser	Adviser dashboard: assigned groups, deliverable statuses, calendar
Student	Student dashboard: my tasks, due dates, announcements, progress

8. Notification System

Superadmin	Configure notification channels; receive platform alerts
Admin	Receive alerts for overdue projects; send broadcast notifications
Adviser	Notified of submissions, new group assignments, and overdue items
Student	Notified of new deliverables, status changes, announcements, deadlines

9. Audit Logs

Superadmin	Full access: all events, all roles, all departments — exportable
Admin	Department-scoped logs: view all actions within their department
Adviser	No access to audit logs
Student	No access to audit logs

9. Notification & Audit Framework

9.1 Automated Notification Triggers

Event Trigger	Notified Roles	Channel
New deliverable added to group	Students, Adviser	In-app + Email
Student submits a deliverable	Research Adviser	In-app + Email
Adviser updates deliverable status	Students (group)	In-app
Deliverable approaching due date (48h)	Students, Adviser	In-app + Email
Deliverable becomes overdue	Students, Adviser, Admin	In-app + Email
New announcement published	All group members	In-app
Admin creates new group or assigns adviser	Adviser, Students	In-app + Email
New Admin account created	Admin	Email (credentials)
New Adviser/Student account created	Adviser/Student	Email (credentials)
Project phase updated	Adviser, Students, Admin	In-app
Suspicious login attempt detected	Superadmin, Admin	Email + Alert

9.2 Audit Log Architecture

All system actions are recorded in an immutable audit log. Each entry captures: Timestamp, User ID & Role, Action Type, Affected Record, IP Address, and Session ID. Audit entries cannot be edited or deleted by any user — including the Superadmin.

✓ User login / logout / failed login attempts	✓ File uploads, downloads, and deletions
✓ Account creation, modification, deactivation, or deletion	✓ Announcement publish/edit/delete actions
✓ Department creation, editing, or archival	✓ Report generation and data exports
✓ Group creation, member changes, adviser assignment	✓ System configuration changes
✓ Deliverable creation, status changes, submissions, approvals	✓ Notification setting changes

Important: Projexis is designed around the principle of least privilege — each role has access only to what is needed to perform their function. This ensures data privacy, accountability, and clean separation of responsibilities across the platform's multi-tenant architecture. Regular review of role assignments and audit logs is strongly recommended for all administrators.